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GEOMAGNETIC POLARITY EPOCHS: NEW DATA FROM OLDUVAI GORGE, TANGANYIKA *

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The lower lava flow of Bed I in Olduvai Gorge, Tanganyika, carries natural remanent magnetization (NRM) having normal polarity. Thermal demagnetization experiments demonstrate the stability of this NRM. Thus the Olduvai geomagnetic polarity event, which was originally named from the upper lava flow in Bed I, is represented in its type locality by two normally magnetized lavas. These lavas have been shown to be 1.9 m.y. old, and although they are distinct from each other in composition and surface structure, their eruptions appear to have been closely spaced in time.

1. INTRODUCTION

Recently an impressive correlation has been shown to exist between radiometric ages of Pleistocene volcanic rocks and the polarities of their natural remanent magnetizations [1-4]. This correlation clearly demonstrates the reality of reversals of the geomagnetic field and has made possible the construction of a time scale of reversals for the last 4 m.y. [1, 3-5]. Time intervals, during which the geomagnetic field has predominantly or entirely one polarity, have been termed epochs, which have durations of the order of 10^6 y., and events, which are about an order of magnitude shorter; thus events may be contained within epochs [1].

In 1963 we reported normal polarity for the NRM of the upper (pahoehoe) basalt flow of Bed I in Olduvai Gorge, Tanganyika, and quoted a potassium-argon age of 1.8 m.y. [6]; this result was not in conflict with the polarity time-scale data available at that time. Later, as it began to appear that the Olduvai result conflicted with other data from rocks of the same age, the suggestion was made [7] that we re-examine this basalt in detail. Subsequently, however, normally magnetized lavas dated at 1.8 to 2.0 m.y. were found on the Pribilof Islands, Alaska [8], leading Cox et al. to conclude that a normal polarity event occurred at that time, which they named the Olduvai event [1]. Further confirmation of the Olduvai event has also been obtained from dated lavas on Réunion [4] and Iceland [9]. A complete account of the potassium-argon dates obtained in Olduvai Gorge has been published by Evernden and Curtis [10], who show that the correct age to be assigned to the basalt of Bed I is 1.9 m.y. This revised age has been quoted by most authors in referring to the paleomagnetic data from Bed I.

The purpose of this paper is to present new paleomagnetic data from the lower (aa) lava flow of Bed I and to show more clearly the relationship between the samples dated by Evernden and Curtis and the samples from which we have obtained paleomagnetic data.

2. DESCRIPTION OF ROCKS

The two lavas of Bed I have been briefly described by Hay [11]. The upper lava is widespread and consists of a series of thin basalt flow units having perfectly preserved pahoehoe crusts; these flow units were apparently extruded over a period of time not exceeding a few tens of years and probably much less. For pa-

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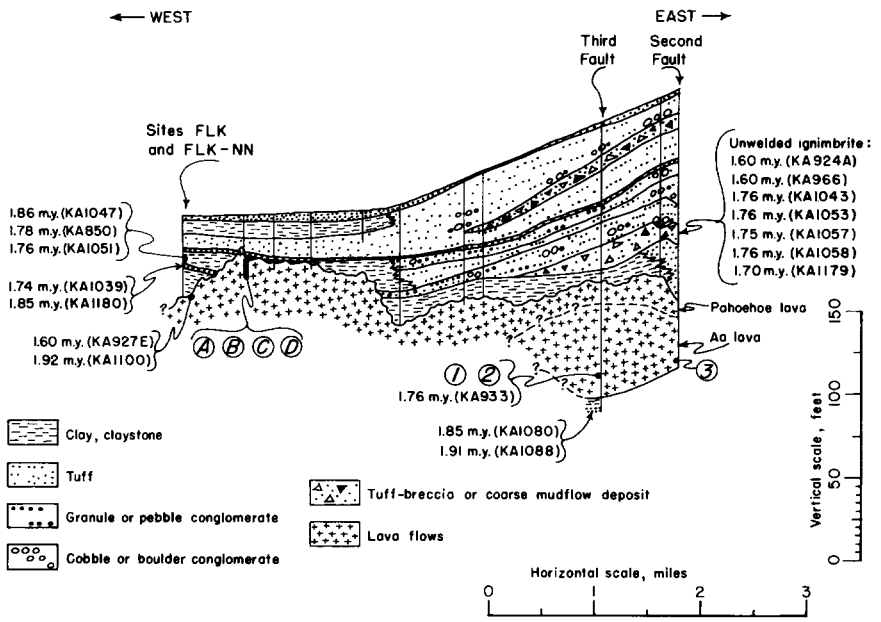


Fig. 1. Stratigraphy of Bed I, Olduvai Gorge, between archeological site FLK and the Second Fault, showing the locations of radiometrically dated samples and paleomagnetic samples. The location of the cross section is shown in fig. 1 of Hay [11]. Vertical lines represent measured stratigraphic sections. The sequence is reconstructed as it would have appeared at the end of Bed I time. Potassium-argon ages and sample numbers are from Evernden and Curtis [10]. Letters A-D represent paleomagnetic samples of pahoehoe basalt [6], and numbers 1-3 represent paleomagnetic samples of aa trachyandesite.

leomagnetic purposes they represent a single point in time and may be referred to as a single lava flow. The lower lava flow is thicker but much less widespread (fig. 1), and has a well-preserved aa surface structure. The pahoehoe lava may have been erupted soon after the aa flow, because the scoriaceous upper surface of the aa flow is unweathered and there is no sedimentary material between the two types of lava.

The petrography of the aa lava has been investigated in two localities: 1) west of the Third Fault, the lava is a slightly to moderately porphyritic mafic trachyandesite. Most of the phenocrysts are augite; the remainder is plagioclase and partly to wholly altered olivine. The groundmass is medium grained and consists largely of plagioclase laths having outer zones of alkalic feldspar (anorthoclase or potassic oligoclase). The remainder of the groundmass consists of clinopyroxene, opaque oxides, altered olivine, an interstitial phyllosilicate mineral and a fraction of 1% of biotite and brown hornblende; 2) west of the Second Fault, the lava is a slightly porphyritic mafic trachyandesite containing phenocrysts of plagioclase, clinopyroxene and partly to wholly altered olivine. The groundmass is fine- to medium-grained and consists largely of plagioclase laths with alkalic outer zones. Next in abundance are clinopyroxene, opaque oxides, and interstitial phyllosilicate material. Also present are a few percent of altered olivine, biotite and pinkish-brown to yellowish-orange hornblende. The lava west of the Second Fault clearly contains more hornblende and biotite than the lava west of the Third Fault. This may indicate that they are actually different flows, but in both areas the aa lava is overlain directly by pahoehoe lava, and if there are two distinct aa flows, they are very likely of virtually the same age.

3. NATURAL REMANENT MAGNETIZATION

Two oriented hand specimens (Nos. 1 and 2) were collected from the aa flow west of the Third Fault, and one specimen (No. 3) was collected from the aa flow west of the Second Fault. Their locations are shown in fig. 1. Three to four 1-in. cores were cut from each hand specimen, and their directions and intensities of magnetization were measured with a "spinner" magnetometer. The NRM directions are shown in fig. 2; the polarity is normal, as for the over-

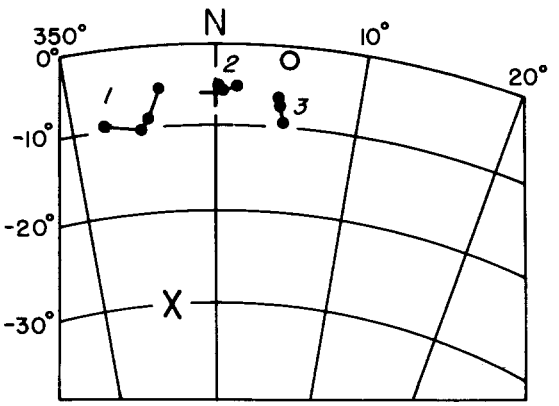


Fig. 2. Equal-area projection on upper hemisphere of directions of magnetization in lower lava flow of Bed I, Olduvai Gorge. Lines connect directions in separate cores cut from the same hand specimens. Also shown is the mean NRM direction in the upper lava flow.
+ Axial dipole field. × Present field.
• North-seeking directions, aa trachyandesite.
o Mean north-seeking direction, pahoehoe basalt.
(From Grommé and Hay [6].)

lying pahoehoe, and the directions are closer to the axial dipole field direction and to the NRM direction in the pahoehoe than to the direction of the present field at that locality. Mean NRM directions and statistical parameters [12] are given in table 1.

A.C. demagnetization was tried on two or three cores from each specimen, but it did not reduce the dispersion of directions. Entries 3 and 4 in table 1 show that the dispersion was greatly increased after demagnetization in only 100 oersteds peak field. This is in marked contrast to our earlier results from the pahoehoe [6], and in general the NRM in the pahoehoe is from two to three times "harder" than that in the aa. Approximately one-third to one-fourth of the original NRM intensity remains in samples of the aa after demagnetization in 200 oersteds.

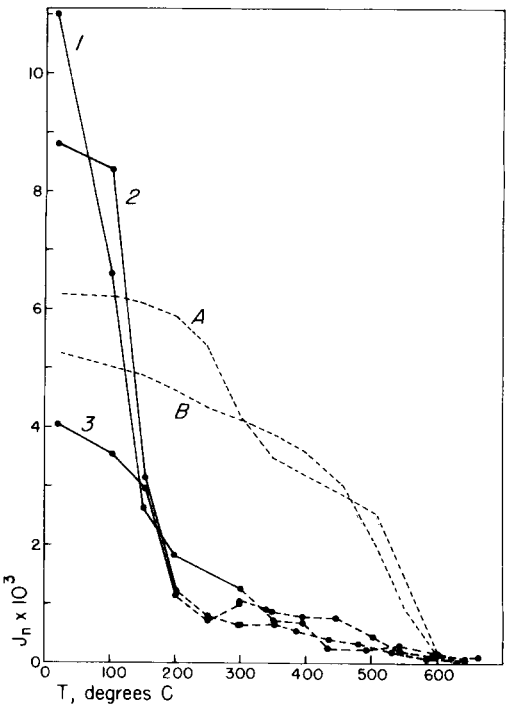


Fig. 3. Thermal decay of intensity J_n of NRM in cores cut from each of three hand specimens from the lower lava flow of Bed I, Olduvai Gorge. Also shown are demagnetization curves (A and B) for the upper lava flow, from Grommé and Hay [6]. (Measurements made at room temperature.)

Thermal demagnetization of NRM in one core from each specimen is illustrated in fig. 3. Two ferromagnetic minerals appear to be present in this lava: one has a Curie temperature between 150° and 200°C and carries most of the NRM; the other shows an apparent distribution of Curie temperatures ranging up to 550°C or more, which may indicate that it has a wide range of compositions. Thermal demagnetization curves previously published for the overlying pahoe-

Table 1
Average north-seeking magnetization directions.

(D = declination east of north; I = inclination, negative upward; J_n = intensity of magnetization in emu/cm^3; other parameters after Fisher [12].)

	Specimen	N	R	k	D	I	α_{95}	J_n
1. Means of cores from hand specimens	1	4	3.9969	955	354.6	-8.6	3.0	9.2×10^{-3}
	2	3	2.9998	8130	0.8	-5.5	1.4	7.7×10^{-3}
	3	3	2.9992	2540	4.6	-7.9	2.5	4.0×10^{-3}
2. Mean of hand specimens		3	2.9915	236	0.0	-7.4	8.1	
3. Mean of all cores		10	9.9662	267	359.4	-7.5		
4. Mean of cores demagnetized in 100 oersteds		7	6.5550	14	(347.0)	(-10.3)		

hoe [6] are also shown for comparison; the ferromagnetic minerals in the two lavas are clearly different.

After heating to temperatures above 200° to 300°C, NRM directions in aa specimens begin to show small random changes, and the intensities of NRM do not decrease smoothly. The corresponding portions of the demagnetization curves are shown dashed in fig. 3. These irregularities result from imperfectly cancelled fields in the non-magnetic oven acting on the low-Curie-temperature fraction of the magnetic minerals during the cooling half of the demagnetizing cycles. In no case was any real tendency toward self-reversal detected, and all of the components of NRM in these specimens are parallel and have the same sense.

The magnetic data neither confirm nor rule out the possibility that the aa lavas exposed at the Second Fault and the Third Fault are parts of different flows. The NRM direction (fig. 2) in specimen 3, from near the Second Fault, differs no more from that in specimen 2 than does the direction in specimen 1, and specimens 1 and 2 were collected close together near the Third Fault. In specimen 3, there appears to be considerably less of the ferromagnetic mineral having a Curie temperature of 150-200°C than in specimens 1 and 2 (fig. 3). This difference, however, is not greater than may be encountered in specimens from widely separated localities on demonstrably the same lava flow, and is hardly more significant than similar differences between specimens A and B from the overlying pahoehoe. Thus for paleomagnetic purposes, we will consider all three specimens of aa lava to represent the same point in time.

4. RELATIONSHIPS OF DATED AND PALEOMAGNETIC SAMPLES

Potassium-argon ages of many rocks from Olduvai Gorge have been published by Evernden and Curtis [10, 13]; these include whole-rock determinations on specimens of both lava flows of Bed I and determinations on minerals separated from tuffs stratigraphically above and below the lavas. The relationships of the various rock units comprising Bed I are shown in fig. 1, together with the locations of paleomagnetic samples and radiometrically dated samples. The potassium-argon ages obtained by Evernden and Curtis may be summarized as follows: a whole-rock determination on a sample from the aa flow (their sample No. KA933) gave an age of 1.76 m.y., which Evernden and Curtis reject as being too young [10]. Samples of the pahoehoe flow (KA927, KA1100) gave ages ranging from 1.60 to 4.4 m.y. [10, 13]; the ages grouped around 4 m.y. are clearly too old, as has already been discussed [11]. An age of 1.92 m.y. determined on a crushed and leached sample of the pahoehoe (KA1100) is regarded by Evernden and Curtis as representing the true age of the lava [10].

Three other groups of age determinations support this age:

1) Feldspars (anorthoclase and plagioclase) separated from tuffs overlying the basalt and associated with the *Zinjanthropus* site gave ages of 1.76, 1.78 and 1.86 m.y. Evernden and Curtis' sample numbers are KA1051, KA850, and KA1047 respectively; KA1051 and KA850 are duplicate determinations on the same sample, but KA1047, which gives the oldest age, is approximately 1 ft higher stratigraphically [10].

2) The most concordant results, and presumably therefore the most reliable dates, were obtained on the unwelded ignimbrite, or *nuée ardente* deposit, that generally lies 3 to 6 ft above the pahoehoe basalt flows (fig. 1). Of the seven age determinations on anorthoclase from this deposit, five range from 1.70 to 1.76 m.y. (KA Nos. 1043, 1053, 1057, 1058 and 1179). The two somewhat discordant dates, both 1.60 m.y. (KA924A, KA966), were obtained from a single sample which also yielded an age of 1.70 m.y. (KA1179) when rerun "by improved procedures" ([10], p. 364). Two determinations made on anorthoclase from a tuff which is probably the ash-fall equivalent of this ignimbrite give ages of 1.74 (KA1039) and 1.85 m.y. (KA1180). The 1.85 m.y. age is considered reliable by Evernden and Curtis [10], although the age of 1.74 m.y. agrees better with dates on the ignimbrite.

3) Anorthoclase and plagioclase were both separated from a tuff exposed in an excavation directly beneath the aa flow near the Third Fault (below the locality from which our samples 1 and 2 were obtained). The anorthoclase gave an age of 1.85 m.y. (KA1080) and the plagioclase an age of 1.91 m.y. (KA1088); both of these determinations are considered reliable by Evernden and Curtis [10].

It has been pointed out by Hay ([10], p. 367) that if an age of 1.75 m.y. for the unwelded ignimbrite is accepted, as seems to be required by the concordance of the ages, then the ages of 1.85 m.y. for KA1180, which is equivalent to the ignimbrite, and 1.86 m.y. for KA1047, which overlies the ignimbrite, must be considered to be about 0.1 m.y. too old. Following this reason-

ing, we can conclude that the ages determined from overlying and underlying rocks bracket the age of the two lava flows of Bed I between 1.75 and 1.9 m.y. If, on the other hand, samples KA1180 and KA1047 were not contaminated by older feldspar and the indicated ages are thus correct, then the age of the lavas is bracketed between 1.85 and 1.9 m.y. In any event, it is clear that the lavas are not older than 1.9 m.y., and if we take the determination on KA1100 (1.92 m.y. from crushed and leached pahoehoe) at face value, then the age of the lavas is 1.9 m.y.

5. SUMMARY AND CONCLUSIONS

Two lavas occur in Bed I in Olduvai Gorge: the upper one is basalt with pahoehoe surface structure and the lower one is mafic trachyandesite with aa surface structure. Potassium-argon age determinations by Evernden and Curtis on a whole-rock sample of the pahoehoe basalt and on feldspars separated from tuffs overlying and underlying the lavas indicate that the age of the lavas is 1.9 m.y. [10].

The polarity of NRM in the pahoehoe basalt had already been reported by us to be normal and to contain two ferromagnetic minerals [6], and Cox et al. [1] concluded that it represented a normal polarity event in the geomagnetic polarity time scale. The aa trachyandesite has now been found to have an NRM direction very similar to that in the pahoehoe. Thermal demagnetization experiments have demonstrated the stability of this NRM. At least two ferromagnetic minerals are present in the rock, and both are magnetized in the same direction and with the same sense. The Olduvai geomagnetic polarity event is thus represented in its type locality by two very different lava flows, whose eruptions occurred 1.9 m.y. ago and were probably closely spaced in time.

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